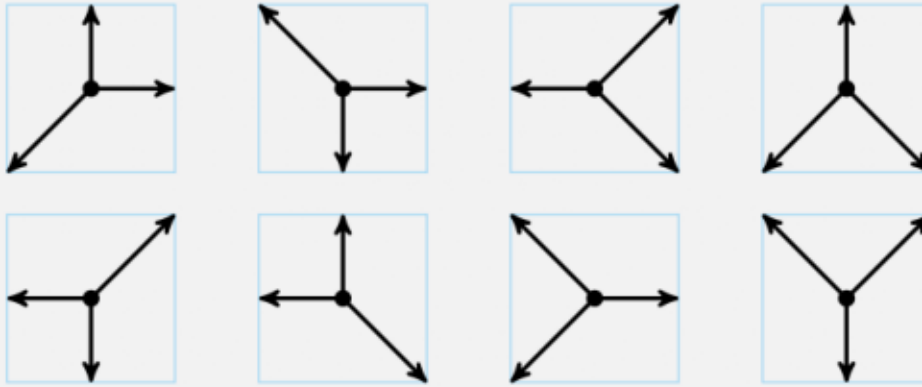


Consider positive spanning sets constructed from the nonzero directions $d_1, d_2 \in \{-1, 0, 1\}$. There are 8 positive spanning sets with 3 elements that can be constructed from these directions:



Example 8.1. Positive spanning sets for $\mathbb{R}^2$. Note that the lower triangular generation strategy can only generate the first two columns of spanning sets.

The rows and columns of $\mathbf{L}$ are then randomly permuted to obtain a matrix $\mathbf{D}$ whose columns correspond to n directions that linearly span $\mathbb{R}^n$. Each of these directions is inscribed inside a hypercube of side length $2/\sqrt{\alpha^{(k)}}$.⁷

Two common methods for obtaining a positive spanning set from the linearly spanning set are to add one additional direction $\mathbf{d}^{(n+1)} = -\sum_{i=1}^n \mathbf{d}^{(i)}$ and to add n additional directions $\mathbf{d}^{(n+j)} = -\mathbf{d}^{(j)}$ for j in $1:n$. We use the first method in algorithm 8.2.

The step size α starts at 1, is always a power of 4, and never exceeds 1. Using a power of 4 causes the maximum possible componentwise step size taken in each iteration to be scaled by a factor of 2, as the maximum componentwise step size $\alpha/\sqrt{\alpha}$ has length $4^m/\sqrt{4^m} = 2^m$ for integer $m < 1$. The step size is updated according to:

$$\alpha^{(k+1)} \leftarrow \begin{cases} \alpha^{(k)} / 4 & \text{if no improvement was found in this iteration} \\ \min(1, 4\alpha^{(k)}) & \text{otherwise} \end{cases} \quad (8.4)$$

Mesh adaptive direct search is opportunistic and does not support dynamic ordering⁸ because, after a successful iteration, the new random pattern may not contain the previous successful direction. The algorithm queries a new design point along the accepted descent direction. If $f(\mathbf{x}^{(k)} = \mathbf{x}^{(k-1)} + \alpha \mathbf{d}) < f(\mathbf{x}^{(k-1)})$,

⁷ For each direction, $\|\mathbf{d}^{(i)}\|_\infty = 1/\sqrt{\alpha^{(k)}}$.

⁸ See section 7.4.